

EXPERIMENT NO. 4

Evaluation of an MCP-Enabled Chatbot

Calculator Tool and Wikipedia Tool using FastMCP, MCP Client-Server Communication and Groq LLM

Post-Lab Evaluation Soft Copy

1. Aim

To evaluate the MCP-enabled chatbot created in Experiment No. 4 by testing it with 20 queries belonging to calculator, Wikipedia, no-tool, mixed and ambiguous categories. The evaluation measures tool selection accuracy, error rate, token usage, MCP communication overhead, unnecessary tool overhead and response time.

2. Theory

Model Context Protocol (MCP)

MCP provides a client-server architecture through which an AI application can discover and invoke external tools. In this experiment, the chatbot acts as the MCP client and connects to a FastMCP server using Streamable HTTP.

FastMCP Server

The server exposes two tools: a calculator tool for mathematical expressions and a Wikipedia search tool for factual, historical and biographical information. Tools are defined on the server using the `@mcp.tool` decorator.

MCP Tool Selection

The LLM receives the available tool descriptions and decides whether a query requires a tool. Calculator queries should use the calculator, factual or historical queries should use Wikipedia search, conceptual queries may require no tool, and mixed queries may require multiple tools.

Evaluation Parameters

The chatbot is evaluated using measured values from 20 test queries. The recorded parameters are tool selection correctness, execution errors, token usage, MCP communication time, unnecessary tool usage and total response time.

3. Project Architecture

Component	Role
User / Evaluation Script	Sends test queries and records evaluation metrics.
Groq LLM Agent	Interprets the query and selects the required tool or answers directly.
MCP Client	Discovers and invokes tools exposed by the MCP server.
FastMCP Server	Hosts the calculator and Wikipedia search tools using Streamable HTTP.
Calculator Tool	Evaluates mathematical expressions.
Wikipedia Search Tool	Retrieves factual, historical and biographical information.
Results Module	Stores per-query results in CSV and generates evaluation graphs.

Flow: User Query → Groq LLM Agent → Tool Selection → MCP Client → FastMCP Server → Calculator / Wikipedia Tool → Tool Result → Final Response → Evaluation CSV and Graphs

4. Twenty Test Queries

No.	Category	Query	Expected Tool(s)
1	Calculator	What is 25 + 75?	calculator
2	Calculator	Calculate 125 * 8	calculator
3	Calculator	What is 1000 / 25?	calculator
4	Calculator	Calculate (45 + 15) * 2	calculator
5	Calculator	What is 2^10?	calculator
6	Wikipedia	Who was Albert Einstein and what were his major contributions?	wikipedia_search
7	Wikipedia	What is the history of the Internet?	wikipedia_search

No.	Category	Query	Expected Tool(s)
8	Wikipedia	Tell me about the history and creator of the Python programming language.	wikipedia_search
9	Wikipedia	Who was Alan Turing and what was his contribution to computer science?	wikipedia_search
10	Wikipedia	What is the history and origin of artificial intelligence?	wikipedia_search
11	No Tool	What is the capital of India?	No Tool
12	No Tool	Explain what machine learning means.	No Tool
13	No Tool	What are the advantages of cloud computing?	No Tool
14	No Tool	What is the difference between AI and ML?	No Tool
15	No Tool	Give me three uses of databases.	No Tool
16	Mixed	Calculate $50 * 20$ and tell me about Alan Turing.	calculator, wikipedia_search
17	Mixed	What is $100 / 4$ and who was Albert Einstein?	calculator, wikipedia_search
18	Ambiguous	Is Python a programming language and when was it created?	wikipedia_search
19	Mixed	Calculate the square of 25 and explain artificial intelligence.	calculator
20	Mixed	Tell me about the Internet and calculate $250 + 750$.	calculator, wikipedia_search

5. Evaluation Results for 20 Queries

No.	Category	Expected	Selected	Correct	Tokens	MCP (s)	Resp. Time (s)	Error
1	Calculator	calculator	calculator	Yes	1831	0.0333	1.3891	No
2	Calculator	calculator	calculator	Yes	1823	0.0317	1.3142	No
3	Calculator	calculator	calculator	Yes	1862	0.0273	2.3191	No
4	Calculator	calculator	calculator	Yes	1874	0.0250	1.1688	No
5	Calculator	calculator	calculator	Yes	1835	0.0269	1.1957	No
6	Wikipedia	wikipedia_search	wikipedia_search	Yes	7218	1.2767	31.4923	No
7	Wikipedia	wikipedia_search	wikipedia_search	Yes	8884	1.3931	64.7289	No
8	Wikipedia	wikipedia_search	wikipedia_search	Yes	13908	16.5771	88.8066	No
9	Wikipedia	wikipedia_search	wikipedia_search	Yes	12166	3.0388	82.6201	No
10	Wikipedia	wikipedia_search	wikipedia_search	Yes	12408	2.7974	72.4025	No
11	No Tool	No Tool	wikipedia_search	No	6020	1.7285	64.9435	No
12	No Tool	No Tool	No Tool	Yes	983	0.0000	0.5792	No
13	No Tool	No Tool	No Tool	Yes	991	0.0000	1.3863	No
14	No Tool	No Tool	No Tool	Yes	993	0.0000	0.8777	No
15	No Tool	No Tool	No Tool	Yes	1006	0.0000	0.6762	No
16	Mixed	calculator, wikipedia_search	calculator, wikipedia_search	Yes	4394	1.2579	3.9661	No
17	Mixed	calculator, wikipedia_search	calculator, wikipedia_search	Yes	3079	0.4311	14.3920	No
18	Ambiguous	wikipedia_search	wikipedia_search	Yes	13919	9.0907	95.3838	No
19	Mixed	calculator	calculator	Yes	2105	0.0988	13.4955	No
20	Mixed	calculator, wikipedia_search	calculator, wikipedia_search	Yes	5731	3.5575	33.9686	No

6. Evaluation of Parameters

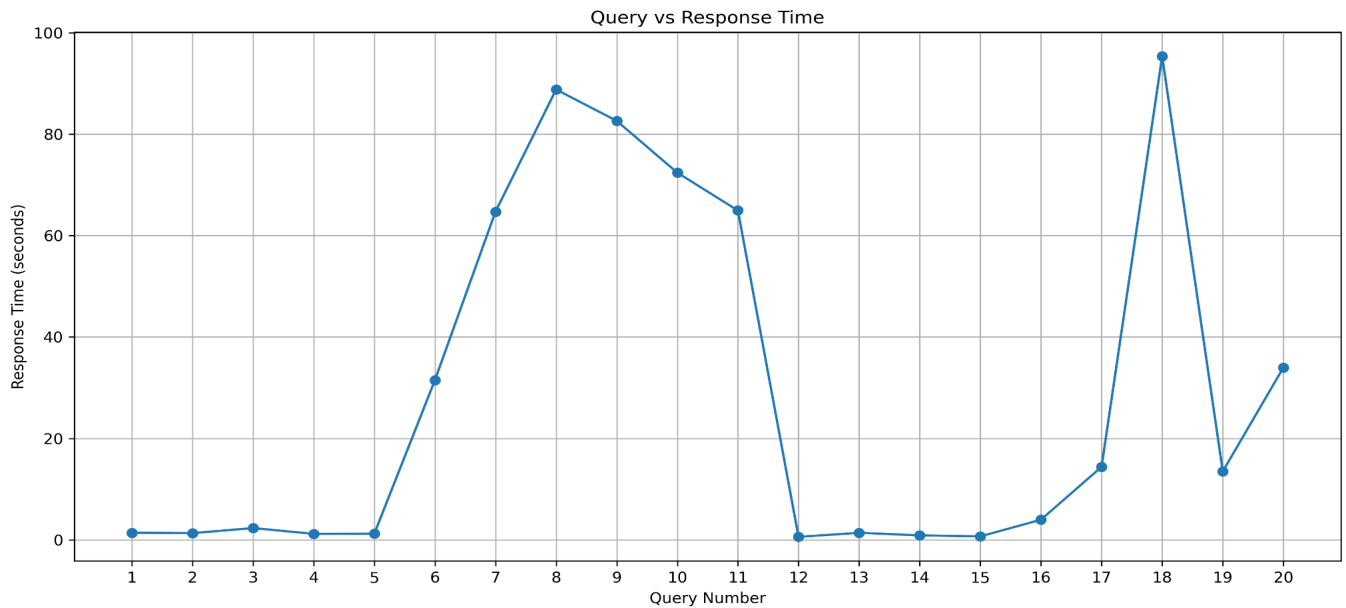
Parameter	Measured Result
Total Queries	20
Tool Selection Accuracy	95.00%
Error Rate	0.00%
Average Token Usage	5151.50 tokens
Average MCP Communication Overhead	2.0696 seconds
Unnecessary Tool Overhead / Rate	5.00%
Average Response Time	28.8553 seconds

Category-wise Tool Selection Accuracy

Category	Accuracy
Calculator	100.00%
Wikipedia	100.00%
No Tool	80.00%
Mixed	100.00%
Ambiguous	100.00%

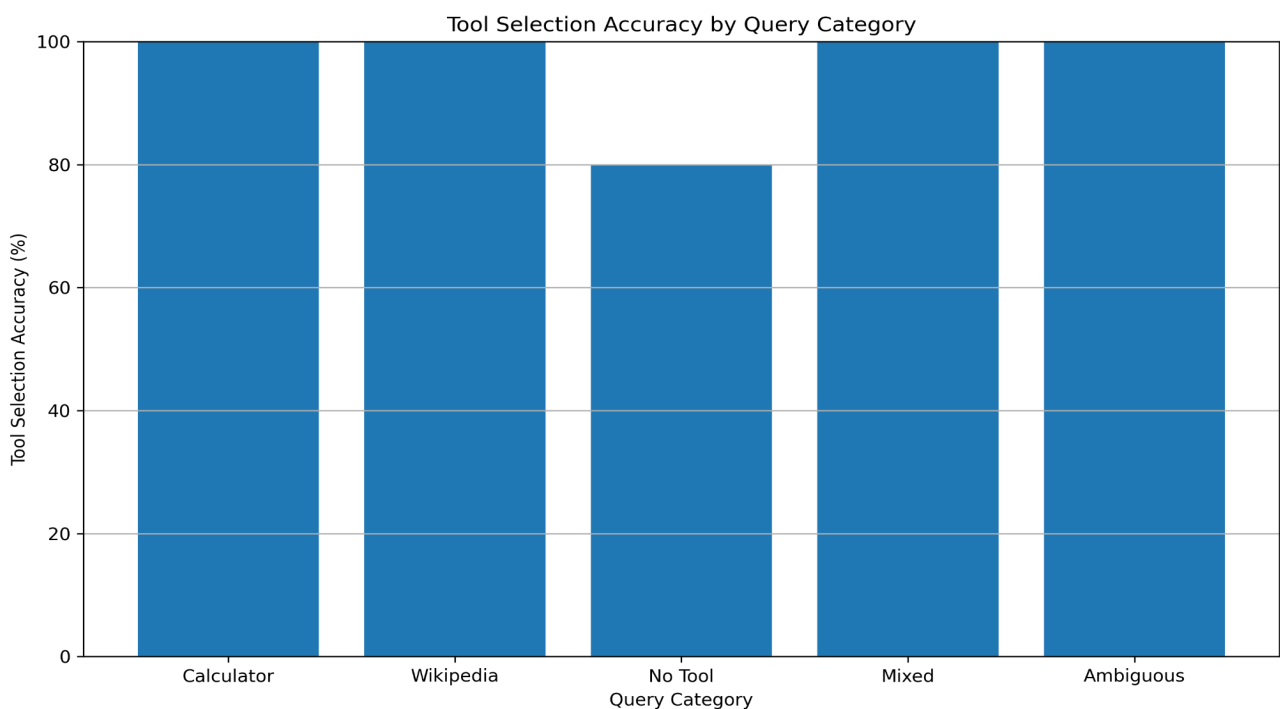
Interpretation: The chatbot correctly selected the required tool for 19 of the 20 evaluated queries. Calculator, Wikipedia, mixed and ambiguous categories achieved 100% measured tool-selection accuracy. The No Tool category achieved 80% because one query, "What is the capital of India?", unnecessarily triggered the Wikipedia tool.

7. Graph 1 — Query vs Response Time



The graph shows the measured response time for each of the 20 test queries. Calculator and no-tool queries generally completed faster, while several Wikipedia and mixed queries required significantly more time because of MCP communication, external information retrieval and larger token processing.

8. Graph 2 — Tool Selection Accuracy by Query Category



The graph represents the measured category-wise tool-selection accuracy. Calculator, Wikipedia, Mixed and Ambiguous categories achieved 100%, while the No Tool category achieved 80%.

9. Final Observations

1. The chatbot completed all 20 evaluation queries without any recorded execution error, resulting in an error rate of 0%.
2. Overall tool selection accuracy was 95%, showing that the agent generally selected the correct MCP tool.

3. All five calculator queries correctly invoked the calculator tool, producing 100% accuracy in the Calculator category.
4. All five Wikipedia queries correctly invoked the Wikipedia search tool, producing 100% accuracy in the Wikipedia category.
5. Mixed and ambiguous queries were handled correctly according to the expected tool combinations, achieving 100% accuracy.
6. One unnecessary Wikipedia tool invocation occurred for the query “What is the capital of India?”, which reduced the No Tool category accuracy to 80% and produced a 5% unnecessary tool rate.
7. The average response time was 28.8553 seconds. Longer response times were mainly observed in queries requiring Wikipedia retrieval and larger token processing.
8. The measured average MCP communication overhead was 2.0696 seconds, indicating the additional time associated with communication between the client and MCP tools.
9. The average total token usage across the evaluation was 5151.50 tokens per query.

10. Conclusion

The MCP-enabled chatbot was successfully evaluated using 20 test queries distributed across calculator, Wikipedia, no-tool, mixed and ambiguous categories. Based on the measured CSV results, the system achieved 95.00% tool selection accuracy with a 0.00% error rate. The experiment demonstrates that MCP allows an LLM-based chatbot to discover and use external tools through a client-server architecture. The calculator and Wikipedia tools were selected correctly in the relevant categories, while one unnecessary tool call showed that further improvement in tool-selection prompts or routing logic could reduce unnecessary overhead. Overall, the experiment successfully demonstrates MCP-based tool integration and quantitative evaluation of chatbot performance.